THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Information Technology and Electrical Engineering
Semester Two Final Examinations, 2022
ELEC7309 Power System Planning and Reliability

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•This is a Closed Book examination - no written materials permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions
•This examination paper will be released to the Library

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

2 x 14-Page Answer Booklet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

There are two Sections in this exam (PART-A and PART-B). Total Marks in the exam is 65%. Answer each section in a separate Booklet.

For Examiner Use Only

Question	Mark

Total _____

PART A: POWER SYSTEM RELIABILITY
ANSWER ALL THREE QUESTIONS FROM PART-A

Question A1 **(Total Marks 10)**

A power system contains three generating units as shown in Table 1.

Table 1 for Question A1: Generation Details

Generator	Capacity (MW)	Failure rate (failure/day)	Repair rate (repair/day)
1	25	0.01	0.49
2	25	0.01	0.49
3	50	0.01	0.49

The system daily peak load over 20 days is recorded in Table 2.

Table 2 for Question A1: Load Profile

Peak load (MW)	Days	Peak load (MW)	Days
65	8	50	4
55	4	46	4

(a) Formulate the capacity outage probability table of the system including both individual outage probability and cumulative outage probability. **(6 marks)**

(b) Calculate the loss of load expectation of the system. (**4 marks)**

Question A2 **(Total Marks 15)**

Three systems are interconnected through two tie lines as shown in Figure 1. The tie line capacity between system A and system B is 15 MW while the tie line capacity between system A and system C is 30 MW. The generation capacity and the load of each system are shown in Table 3 below. Assume the tie lines are 100% reliable and the load duration curve of each system during one year period is a straight line from the 100% to 75% load points. The interconnection agreement among the three systems is “assist if possible”.

(a) Calculate the loss of load expectation of system A without the interconnection. **(5 Marks)**

(b) What is the loss of load expectation for system A when it is interconnected with system B and system C? **(10 Marks)**

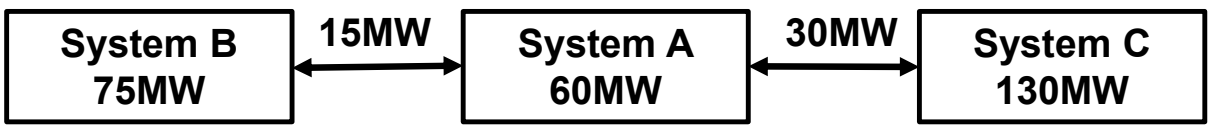


Figure 1 for Question A2

Table 3 for Question A2: Generation Details

System	Generation unit	Generation capacity (MW)	Force outage rate of each generator	Total generation capacity (MW)	Peak load (MW)
A	4	10	0.02	60	40
	1	20	0.02		
B	5	10	0.02	75	50
	1	25	0.02		
C	5	20	0.02	130	90
	1	30	0.02		

Question A3 **(Total Marks 10)**

A radial single-phase power system is shown in Figure 2 which contains only one circuit breaker. Assume that the failure rate of the lines is approximately proportional to their length. For example, the failure rate of the main feeder (sections 1, 2, 3, and 4) is 0.1 f/km/yr and the failure rate of the lateral distribution lines (a, b, c, and d) is 0.25 f/km/yr. Assume that the repair time *r* for each of the main feeder is 3 hours and each of the lateral distribution lines is 1 hour.

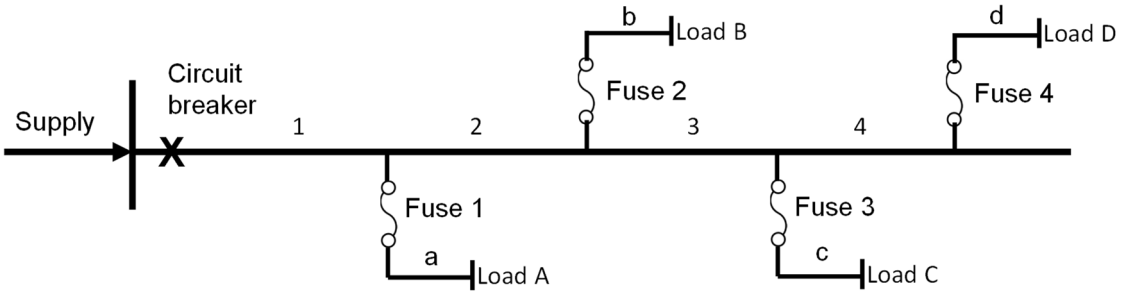


Figure 2 for Question A3

Reliability parameters for the system of Figure 2 is shown in Table 4.

Table 4 for Question A3: Reliability Parameters

Component	Length (km)	Component	Length (km)
Section- 1	2	Distributor -a	1
Section- 2	2	Distributor -b	1
Section- 3	1	Distributor -c	1
Section- 4	1	Distributor -d	1

Customers and load connected to the system in Figure 2 is shown in Table 5.

Table 5 for Question A3: Customers and Load Details

Load point	Number of customers	Average load connected (kW)
A	300	3000
B	250	4000
C	200	2500
D	150	2000

Assume that short circuits happen to each component and each failure on the main feeder will trip the main circuit breaker. A fuse is installed at the tee-point in each lateral distribution line which will blow when a short circuit happens on a lateral distributor. On this basis, what would be the reliability indices of the distribution system including SAIFI, SAIDI, CAIDI, ASUI and ENS?

PART B: POWER SYSTEM PLANNING
ANSWER All FIVE QUESTIONS FROM PART-B

Question B1 **(Total Marks 6)**

Consider the three investment options detailed in Table 6 for economic evaluation. The generators of Options 1 and 3 are each 9 MW. Option 1 (Gas Generator) operates at full capacity (100% capacity factor) while Option 3 (Solar Generator) operates at 32% capacity factor. Implementation of each option will result in system loss reductions and the reductions in expected energy not supplied (EENS) are detailed in Table 7.
In undertaking the economic evaluation, assume 100% reliability and the annual weighted cost of marginal production of the existing generating plant is $24/MWh. Also, assume that the VCR (value of customer reliability) is $36/kWh. Consider discount rate of 5% per annum.
Estimate the net present value (NPV) for each of the three options and determine the preferred option (highest NPV).

Table 6 for Question B1: Investment Option Details

Options	Capital Cost	Fuel Cost	Maintenance Cost	Assets Life
Option 1 - 9 MW Gas Generator	$10,000,000	$15/MWh	$410,000 p.a.	20 years
Option 2 – Transmission Line	$5,000,000	0	$50,000 p.a.	30 years
Option 3 – 9 MW Solar Generator	$18,000,000	0	$300,000 p.a.	20 years

Table 7 for Question B1: Benefit Details

Options	System Loss Reduction	Reduced EENS
Option 1 - 9 MW Gas Generator	420 MWh	50 MWh
Option 2 – Transmission Line	1,100 MWh	40 MWh
Option 3 – 9 MW Solar Generator	2,000 MWh	45 MWh

Question B2 **(Total Marks 6)**

An expansion project has two plans (Plan A and Plan B) as outlined in Table 8. Assume, both plans have equal benefits.

a) Using the Least Cost method for economic evaluation, determine the better plan in consideration of Present Value of Cost (PVC) for the expansion project. **(4 marks)**

b) Using the Constant Annuity method for economic evaluation, determine the better plan for the expansion project. **(2 marks)**

Table 8 for Question B2

Expansion Project	Plan A	Plan B
Investment	\$2000k	\$4000k
Residual Value (RV)	0	\$800k
Lifetime	3 years	4 years
Annual cost	\$700k	\$500k
Discount rate	5%	5%

Question B3 **(Total Marks 5)**

The marginal costs of generations for two systems (System X and System Y), shown in Figure 3, are governed by the following two equations (with P_X and P_Y are in MW).

$$MC_X = 18 + 0.05 \cdot P_X \text{ (in \$/MWh)}$$
$$MC_Y = 16 + 0.04 \cdot P_Y \text{ (in \$/MWh)}$$

Consider both systems of Figure 3 are operated with the marginal cost functions shown in the equations above, and they are interconnected without a limit on transfer capacity. Calculate the electricity price and the electricity cost per hour for the customers in each system; calculate the cost of generation production per hour, the revenue per hour of the generator, and the profit made by the generator per hour in each system.

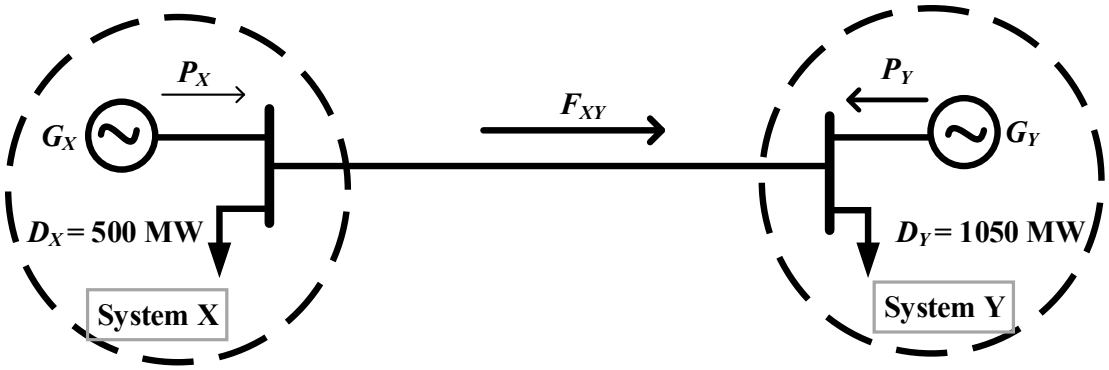


Figure 3 for Question B3: System Details

Question B4 **(Total Marks 5)**

Two zones (Zone A and Zone B) are interconnected through a transmission line, see Figure 4(a). The Load Duration Curves for Zone A and Zone B are illustrated in Figure 4(b). The marginal costs of generations for two zones are governed by the following two equations (with P_A and P_B are in MW).

$$MC_A = 11 + 0.01 \cdot P_A \text{ (in \$/MWh)}$$
$$MC_B = 14 + 0.02 \cdot P_B \text{ (in \$/MWh)}$$

Now, if the interconnection capacity is constrained to 200 MW, compute the hourly generation cost (total) and hourly constraint cost to meet the off-peak and on-peak load demand. Also, calculate the yearly constraint cost (total).

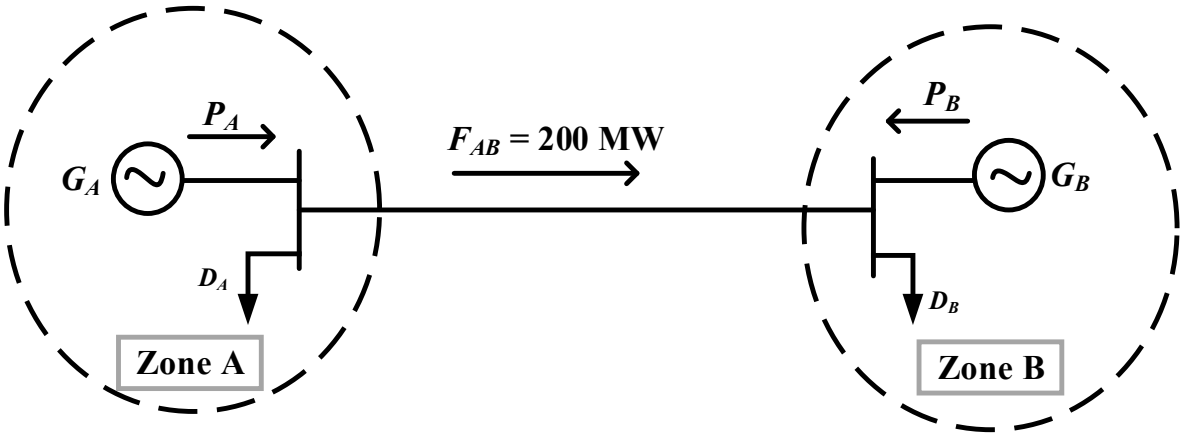


Figure 4(a) for Question B4

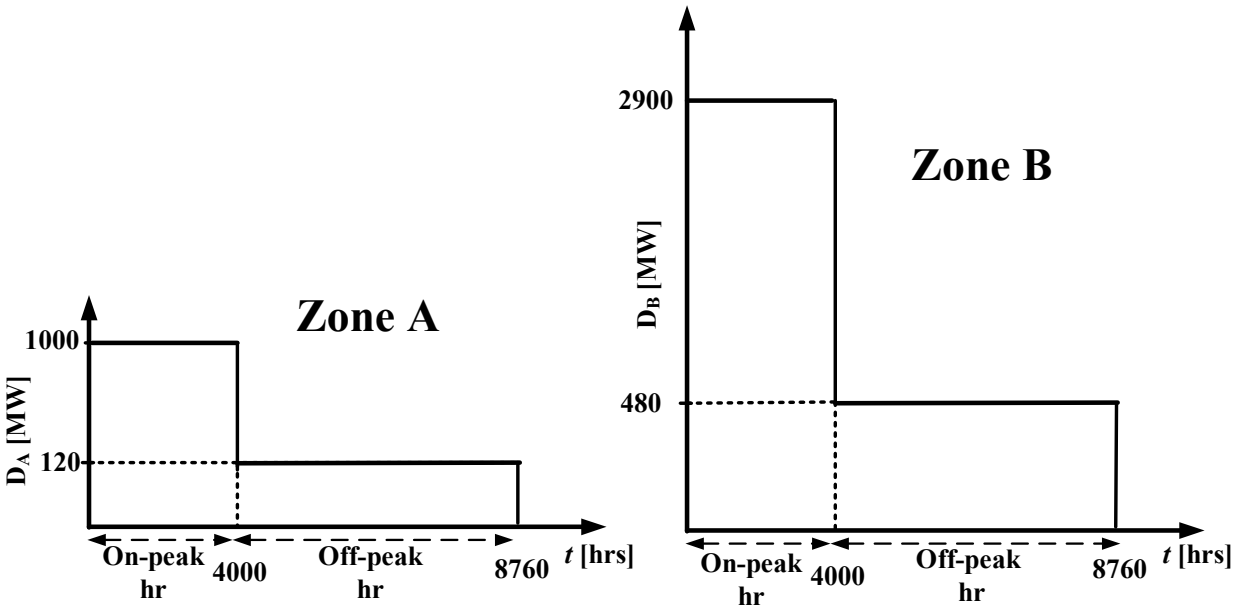


Figure 4(b) for Question B4: Load Duration Curves

Question B5 **(Total Marks 8)**

For three generation facilities, X, Y and Z, with details given in Table 9, assume an interest rate of 8% and the possibility of choosing any generation capacity of the above-mentioned technologies (X, Y and Z).

a) Derive the equations of annual cost as a function of capacity factor for each of the technologies using the data given in Table 9 and the Load Duration Curve shown in Figure 5. **(3 Marks)**

b) Using the Screening Curve technique, determine the optimal type and capacity to supply the demand characterised by the Load Duration Curve shown in Figure 5. **(5 Marks)**

Table 9 for Question B5: Generation Technology Data

Technology type	Investment cost ($/kW)	Plant life (years)	Fuel cost ($/MWh)
X	1000	34	18
Y	380	29	31
Z	90	24	49

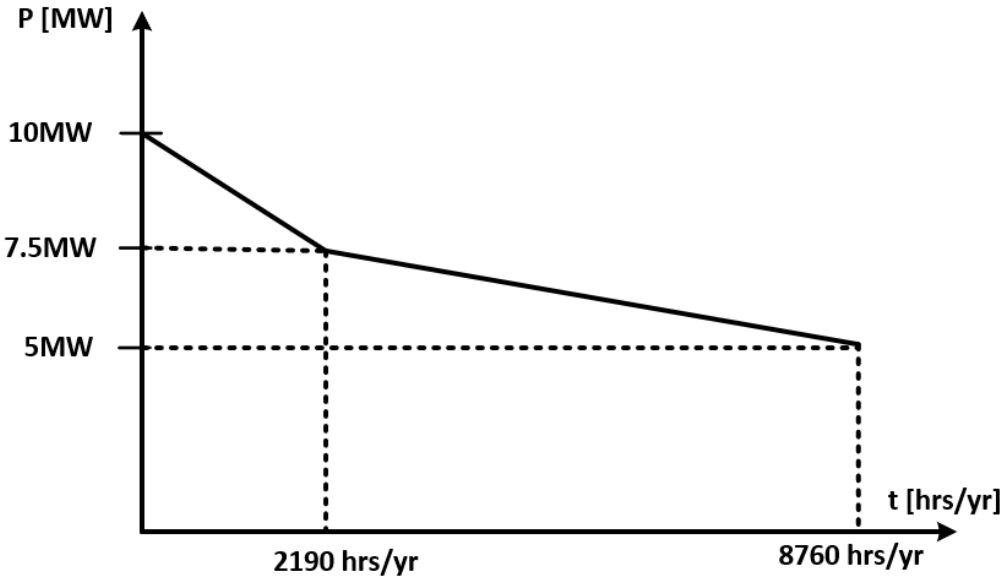


Figure 5 for Question B5: Load Duration Curve

END OF EXAMINATION

Formulae-PART A

General

Failure rate/Forced Outage Rate: λ

Repair Rate: μ

Mean Time Before Failure (MTBF) $\frac{1}{\lambda}$

Mean Time To Repair (MTTR): $r = \frac{1}{\mu}$

For Generation and Composite Power System Reliability Assessment

Availability: $A = \frac{\mu}{\mu+\lambda}$

Unavailability: $U = \frac{\lambda}{\mu+\lambda}$

For Distribution Reliability Assessment

Availability: $A = 1$

Unavailability: $B = \lambda \cdot r = \frac{\lambda}{\mu}$

Common Customer Reliability Indices

System Average Interruption Frequency Index

$$\text{SAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers served}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

System Average Interruption Duration Index

$$\text{SAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customers served}} = \frac{\sum N_i \cdot Q_i}{\sum N_i}$$

Customer Average Interruption Frequency Index

$$\text{CAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers affected}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

Customer Average Interruption Duration Index

$$\text{CAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customer interruptions}} = \frac{\sum N_i \cdot Q_i}{\sum N_i \cdot F_i}$$

Average Service Availability Index

$$\text{ASAI} = \frac{\text{customer hours of available service}}{\text{customer hours demanded}} = \frac{\sum N_i \cdot (1 - Q_i)}{\sum N_i}$$

Average Service Unavailability Index $\text{ASUI} = 1 - \text{ASAI}$

Energy Not Supplied

$$\text{ENS} = \sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}$$

Average Energy Not Supplied

$$\text{AENS} = \frac{\text{total energy not supplied}}{\text{numbers of customers served}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$

Average Curtailment Index

$$\text{ACCI} = \frac{\text{total energy not supplied}}{\text{number of customers affected}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$

Composite System Reliability Analysis

If the occurrence of an event A is dependent upon a number of event B_i which are mutually exclusive, then the system failure event A will be;

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i) P(B_i)$$

$$P_k = \sum_j P(B_j)(P_{gj} + P_{lj} - P_{gj} P_{lj})$$

$$F_k = \sum_j [F(B_j)(P_{gj} + P_{lj} - P_{gj} P_{lj})$$

Here;

B_j= outage state j in the transmission network

P_{gj} = probability of generating capacity outage at state j exceeding the reserve margin

P_{lj} = probability of state j at load bus k exceeding the maximum load that can be supplied at that bus without failure.

P_c = probability of transmission inadequacy

Where $F(B_j)$ is the frequency of occurrence of outage B_j

$$F_C = \sum_{j=1}^{2^m} \sum_{i=1;i\neq j}^{2^m} \underbrace{(1 - P_C(j))} \cdot \underbrace{P_C(i)} \cdot \lambda_{ji} \qquad P_C = \sum_{j=1}^{2^m} P(j) \cdot P_C(j)$$

$$F = \underbrace{(1 - P_G)} \cdot F_C + \underbrace{(1 - P_C)} \cdot F_G$$

Recursive algorithm for capacity model building

$$p(X) = p'(X)(1-U) + p'(X-C)U$$

$$\lambda_+(X) = \frac{p'(X)(1-U)\lambda_+(X) + p'(X-C)U(\lambda'_+(X-C) + \mu)}{p(X)}$$

$$\lambda_-(X) = \frac{p'(X)(1-U)(\lambda'_-(X) + \lambda) + p'(X-C)U(\lambda'_-(X-C))}{p(X)}$$

$$P(X) = P(Y) + p(X)$$

$$F(X) = F(Y) + p(X)(\lambda_+(X) - \lambda_-(X))$$

Here;

p (X), λ_+ (X) and λ_- (X) are the individual state probability and the upward and downward capacity departure rates respectively after the unit is added

Y denotes the capacity outage state just larger than X MW

For X < C ; p'(X - C) = 0, λ'_+(X-C) = 0 and λ'_-(X-C) = 0

For X ≠ 0 , C_1, the procedure is initiated with the addition of the first unit (C_1), in this case; $\lambda_+(0) = 0$, $\lambda_-(0) = \lambda_1$, $\lambda_+(C_1) = \mu_1$, $\lambda_-(C_1) = 0$, $\lambda_+(X) = \lambda_-(X) = 0$

Formulae-PART B

Variable names and meaning

- P_0: present value
- F_n: value in n years
- A: annuity (constant)
- A_n: annuity in year n
- i: interest rate
- B_{jt}: profit of plan j in year t
- C_{jt}: operation cost of plan j in year t
- K_{jt}: investment of plan j in year t
- j: number of the plan in consideration
- n_j: horizon of study (service/lifetime) of plan j

Present value

$$P_0 = \frac{F_n}{(1+i)^n}$$

Present Value Factor $(P|F,i,n)$:

$$(P|F,i,n) = \frac{1}{(1+i)^n}$$

Future Value

$$F_n = (1+i)^n \cdot P_0$$

Future Value Factor $(F|P,i,n)$:

$$(F|P,i,n) = (1+i)^n$$

Constant Annuity

$$F_n = A\frac{(1+i)^n - 1}{i} = A(F|A,i,n)$$
$$A = F_n\frac{i}{(1+i)^n - 1} = F_n(A|F,i,n)$$

$$P_0 = \frac{F_n}{(1+i)^n} = A\frac{(1+i)^n - 1}{i\cdot(1+i)^n} = A\frac{1-(1+i)^{-n}}{i} = A(P|A,i,n)$$
$$A = P_0\frac{i\cdot(1+i)^n}{(1+i)^n - 1} = P_0\frac{i}{1-(1+i)^{-n}} = P_0(A|P,i,n)$$

Capital Recovery Factor

$$CRF_j = \frac{i\cdot(1+i)^{n_j}}{(1+i)^{n_j} - 1} = \frac{i}{1-(1+i)^{-n_j}} = (A|P,i,n)$$

Incremental Annuity

$$P_0 = \sum_{k=1}^{n} A_k(1+i\,)^{-k} = \sum_{k=1}^{n} A_k(P|F,i,k)$$

Net present value

$$NPV_j = \sum_{t=1}^{n_j}\left[B_{jt}(P|F,i,t)\right] - \sum_{t=1}^{n_j}\left[\left(C_{jt}+K_{jt}\right)(P|F,i,t)\right]$$

Benefit/Cost Ratio

$$BCR_j = \frac{\sum_{t=1}^{n_j}\left[\left(B_{jt}-C_{jt}\right)(P|F,i,t)\right]}{\sum_{t=1}^{n_j}\left[K_{jt}(P|F,i,t)\right]}$$

Present Value of Costs

$$PVC_j = \sum_{t=1}^{n_j}\left[\left(C_{jt}+K_{jt}\right)(P|F,i,t)\right]$$